

EXAM 7 – SPRING 2015

3. (2.25 points)

Given the following information:

Cumulative Reported Claims (\$000)

<u>Accident Year</u>	<u>As of 12 Months</u>	<u>As of 24 Months</u>
2008	1,700	3,600
2009	2,300	3,200
2010	1,200	1,700
2011	500	2,600
2012	2,600	3,000
2013	700	2,100

a. (1.5 points)

Calculate the weighted residuals for reported claims as of 24 months following Mack's methodology. Create a scatter plot of these residuals against cumulative reported claims as of 12 months.

b. (0.75 point)

Identify which one of the three assumptions specified by Mack as underlying the chain ladder method can be tested by reviewing the scatter plot created in part a. above, and discuss whether the assumption has been violated.

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QUESTION 3

TOTAL POINT VALUE: 2.25

LEARNING OBJECTIVE: A2: Estimate parameters and unpaid claims using claims development models related to loss reserving methods such as chain ladder, Cape Cod, chain ladder plus calendar-year effects, and Bornhuetter-Ferguson.

SAMPLE ANSWERS

Part a: 1.5 points

Sample Answer 1

$$\text{Weighted residual} = \frac{C_{i,k+1} - C_{i,k} \cdot f_k}{\sqrt{C_{i,k}}}$$

$$f_k = \text{Sum of 24 month cumulative} / \text{sum of 12 month cumulative} = 16200 / 9000 = 1.8$$

AY	Cik	Ci k+1	fCik	Residual
08	1700	3600	3060	13.097
09	2300	3200	4140	(19.6)
10	1200	1700	2160	(13.279)
11	500	2600	900	76.026
12	2600	3000	4680	(32.95)
13	700	2100	1260	31.75

Sample Answer 2

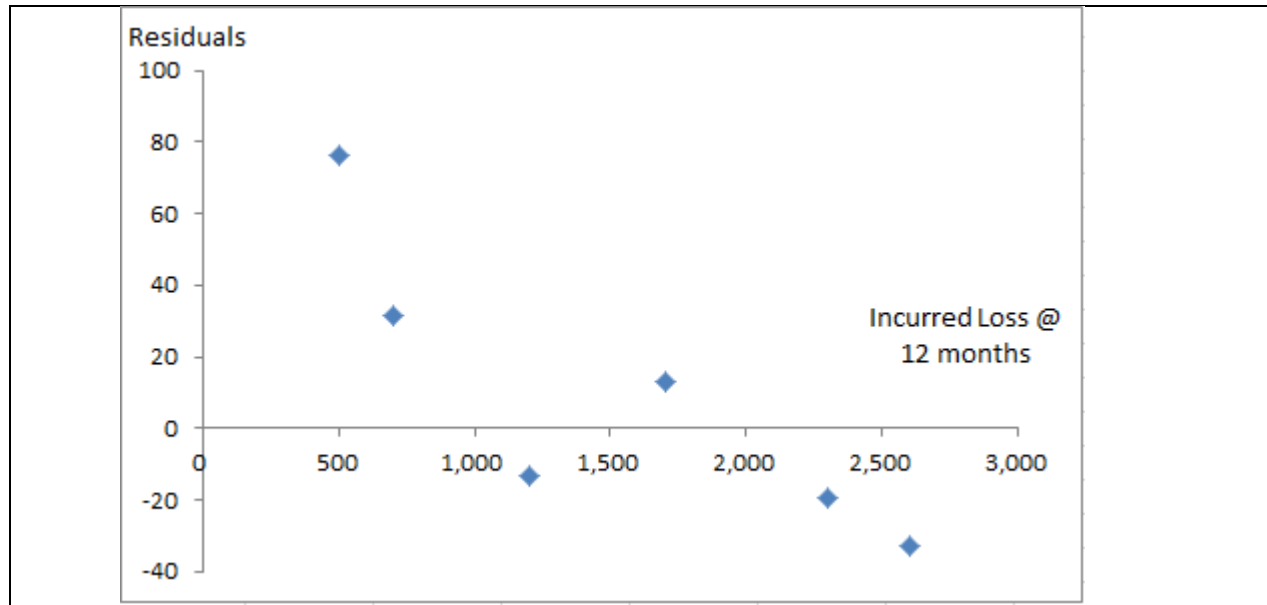
In full dollars				
AY	12	A-24	E-24	R
2008	1,700,000	3,600,000	3,060,000	414.161
2009	2,300,000	3,200,000	4,140,000	-619.818
2010	1,200,000	1,700,000	2,160,000	-419.921
2011	500,000	2,600,000	900,000	2404.163
2012	2,600,000	3,000,000	4,680,000	-1041.892
2013	700,000	2,100,000	1,260,000	1003.992

Sample Answer 3

In millions				
AY	12	A-24	E-24	R
2008	1.7	3.6	3.1	0.414
2009	2.3	3.2	4.1	-0.620
2010	1.2	1.7	2.2	-0.420
2011	0.5	2.6	0.9	2.404
2012	2.6	3.0	4.7	-1.042
2013	0.7	2.1	1.3	1.004

Sample Residual Plot

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Part b: 0.75 point

Sample Answer 1

This is testing the assumption that the variance of the next period's losses is proportional to the prior periods reported loss. For the assumption to be met, we expect to see the residuals randomly scattered around 0. This is not the case with this plot as we clearly see a decreasing trend in the residuals as claim size increases. The assumption has not been met.

Sample Answer 2

Variance of next year's incurred loss is proportional to incurred loss to date and a factor based on age. Since the points have a decreasing pattern (i.e. not random), the assumption is violated.

EXAMINERS' REPORT

Part a

- Candidates were expected to know how to calculate weighted residuals and weighted loss development factors using Mack's method.
- Candidates generally scored well on this part.
- Common mistakes encountered were inability to recall formulas, simple computational errors, and not appropriately labeling the graph axes.

Part b

- Candidates were expected to know the relation of the variance assumptions to methods of calculating development factors and how to test whether these assumptions have been violated or not.
- The most common mistake was to refer to the expected value rather than variance assumption.